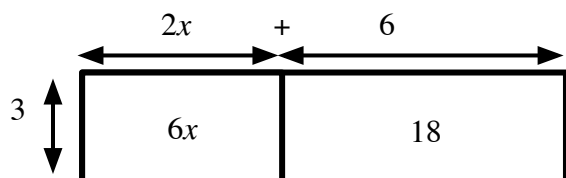




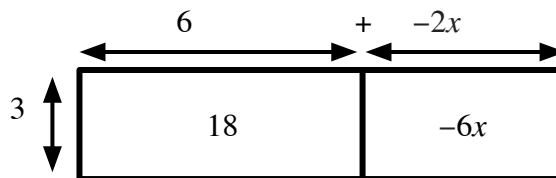
Checking and securing understanding of the distributive law with algebraic terms

Task A: Multiplying expressions by a term

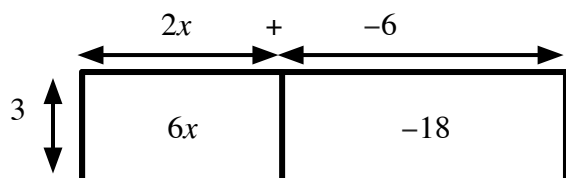
1) Use the area models to write these expressions in expanded form



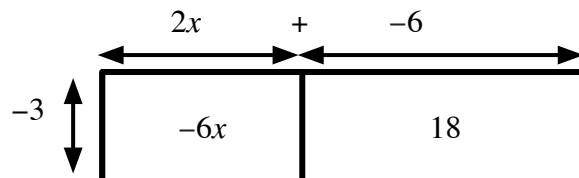
a) $3(2x + 6) \equiv 6x + 18$



c) $3(6 - 2x) \equiv 18 - 6x \equiv -6x + 18$

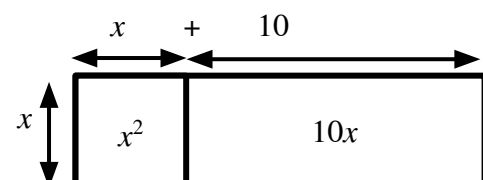


b) $3(2x - 6) \equiv 6x - 18$

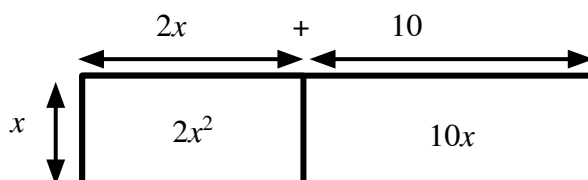


d) $-3(2x - 6) \equiv -6x + 18 \equiv 18 - 6x$

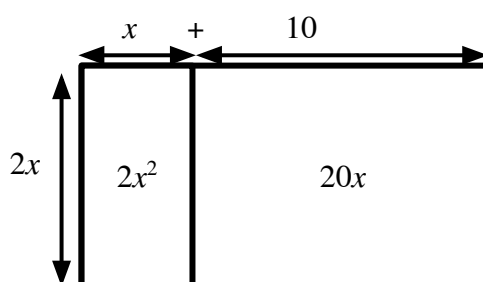
2) Use the area models to write these expressions in expanded form



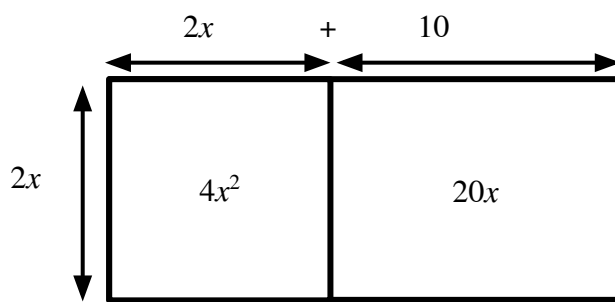
a) $x(x + 10) \equiv x^2 + 10x$



c) $x(2x + 10) \equiv 2x^2 + 10x$



b) $2x(x + 10) \equiv 2x^2 + 20x$



d) $2x(2x + 10) \equiv 4x^2 + 20x$

3) Write each of these in expanded form.

a) $5(x + 4) \equiv 5x + 20$

h) $5x(x + 4) \equiv 5x^2 + 20x$

b) $5(2x + 4) \equiv 10x + 20$

i) $5x(2x + 4) \equiv 10x^2 + 20x$

c) $5(2x - 4) \equiv 10x - 20$

j) $5x(2 + 4x) \equiv 10x + 20x^2$

d) $10(2x - 4) \equiv 20x - 40$

k) $5x(2 - 4x) \equiv 10x - 20x^2$

e) $10(4 - 2x) \equiv 40 - 20x$

l) $-5x(2 - 4x) \equiv -10x + 20x^2$

f) $10(-2x + 4) \equiv 40 - 20x \equiv -20x + 40$

m) $-5x(2 - 4x + y) \equiv -10x + 20x^2 - 5xy$

g) $-10(-2x + 4) \equiv 20x - 40$

n) $-5x(2 - 4x) + y \equiv -10x + 20x^2 + y$

Task B: Multiplying and simplifying multiple expressions

1) Expand and simplify where possible.

a) $4(2x + 6) + 5(x + 2)$

$$8x + 24 + 5x + 10$$

$$13x + 34$$

b) $7(x - 5) + 2(x + 13)$

$$7x - 35 + 2x + 26$$

$$9x - 9$$

c) $x(2x + 1) + 6(5x - 3)$

$$2x^2 + x + 30x - 18$$

$$2x^2 + 31x - 18$$

d) $-5x(3x + 7) - x(3x + 7)$

$$15x^2 + 35x - 3x^2 - 7x$$

$$12x^2 + 28x$$

e) $4x(2y + 7) - 2(3x - 4)$

$$8xy + 28x - 6x + 8$$

$$8xy + 22x + 8$$

f) $-3(4y + 7x) - 2(4y + 7x)$

$$4y + 7x$$

g) $-5x(8y - 7x) - (yx - 10x)$

$$-40xy + 35x^2 - yx - (-10x)$$

$$-40xy + 35x^2 - yx + 10x$$

$$35x^2 - 41xy + 10x$$

2) All four of these expressions are equivalent to each other.

Fill in the missing terms.

a) $7(x - 3) + 4(2x + 6.5) \equiv 7x - 21 + 8x + 26 \equiv 15x + 5$

b) $2(3x + 4) + 3(3x + \boxed{-1}) \equiv 6x + 8 + 9x + ? \quad ? = -3$

c) $5(\boxed{4x} - 1) - 5(x - 2) \equiv ? - 5 - 5x + 10 \quad ? = 20x$

d) $7(3x + 1) - \boxed{2}(3x + 1) \equiv 15x + 5 \equiv 5(3x + 1)$

